# QIUZHEN QUALIFY EXAM FOR ARTIFICIAL INTELLIGENCE

## MOCK EXAMINATION NO. 1

### Instructions

- This examination consists of four sections, each worth 33 points. Choose exactly three sections and answer all questions in the selected sections.
- Write your name and student ID on the answer sheet for 1 additional point. The maximum possible score is 100.
- State the three selected sections at the beginning of the answer sheet. If more than three sections are attempted, only the first three will be graded.
- Unless stated otherwise, justify every claim and show the essential derivation. Unsupported final formulas receive limited credit.

---

## A. Machine Learning Theory

**1. [3 pts.]** For each item, select exactly one answer. No justification is required. Each item is worth 0.5 points.

(i) In the realizable PAC setting, finite VC dimension is sufficient for ERM to be a PAC learner. (True / False)

(ii) A symmetric function $K$ is a valid kernel whenever $K(x, x) \geq 0$ for all $x$. (True / False)

(iii) Unregularized logistic regression on strictly linearly separable data always has a finite maximum-likelihood estimator. (True / False)

(iv) For every $\lambda > 0$, ridge regression has a unique coefficient vector even if the design matrix is rank deficient. (True / False)

(v) In agnostic PAC learning, the learner competes with the best hypothesis in the prescribed class. (True / False)

(vi) A larger training accuracy by itself implies a smaller population risk. (True / False)

**2. [7 pts.]** For $a = (a_1, \ldots, a_d) \in \mathbb{R}^d$, define

$$h_a(x) = \mathbf{1}\{x_j \leq a_j \text{ for every } j = 1, \ldots, d\}, \qquad x \in \mathbb{R}^d.$$

Determine the VC dimension of $\mathcal{H} = \{h_a : a \in \mathbb{R}^d\}$. Prove both the lower and upper bounds.

**3. [8 pts.]** Let $z_i \in \mathbb{R}^p$ be augmented feature vectors (including any intercept coordinate) and $y_i \in \{-1, +1\}$. For $\lambda > 0$, consider

$$J(\theta) = \frac{1}{m} \sum_{i=1}^{m} \log\left(1 + e^{-y_i \theta^\top z_i}\right) + \frac{\lambda}{2} \|\theta\|_2^2 .$$

(a) **[3 pts.]** Derive $\nabla J(\theta)$.

(b) **[3 pts.]** Derive $\nabla^2 J(\theta)$ and prove that $J$ has a unique minimizer.

(c) **[2 pts.]** Set $\lambda = 0$. If the data are strictly linearly separable, determine whether a finite minimizer must exist and justify the answer.

**4. [9 pts.]** Consider the soft-margin support vector machine

$$\min_{w,b,\xi} \frac{1}{2}\|w\|_2^2 + C\sum_{i=1}^{m}\xi_i \quad \text{s.t.} \quad y_i(w^\top x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0,$$

where $y_i \in \{-1, +1\}$ and $C > 0$.

(a) **[3 pts.]** Form the Lagrangian with the correct multiplier signs and eliminate the primal variables.

(b) **[3 pts.]** State the dual problem, including every constraint.

(c) **[3 pts.]** State the complementary-slackness conditions. Explain how to recover $b$ from any training point whose dual multiplier satisfies $0 < \alpha_i < C$, and identify which points contribute to the classifier.

**5. [6 pts.]** Let $\mathcal{X} = \{1, \ldots, N\}$ and

$$K_c(i, j) = \min\{i, j\} - c.$$

Determine all real $c$ for which $K_c$ is a positive-semidefinite kernel on $\mathcal{X}$. For the valid range, give an explicit finite-dimensional feature map; for the invalid range, give a direct certificate of failure.

## B. Deep Learning and Reinforcement Learning

**1. [8 pts.]** For one example $x \in \mathbb{R}^d$ with label $y \in \{1, \ldots, C\}$, consider

$$a = W_1 x + b_1, \quad h = \text{ReLU}(a), \quad o = W_2 h + b_2, \quad p = \text{softmax}(o), \quad \ell = -\log p_y,$$

where $W_1 \in \mathbb{R}^{r\times d}$ and $W_2 \in \mathbb{R}^{C\times r}$. Derive the gradients with respect to $o, W_2, b_2, a, W_1, b_1$. State the dimension of every gradient and specify how a ReLU derivative is chosen at zero.

**2. [8 pts.]** Let $X \in \mathbb{R}^{L\times d}$ and define a single-head self-attention layer by

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V, \qquad Y = \text{softmax}_{\text{row}}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V,$$

where $W_Q, W_K \in \mathbb{R}^{d\times d_k}$ and $W_V \in \mathbb{R}^{d\times d_v}$.

(a) **[2 pts.]** Give the dimensions of $Q, K, V$, the attention matrix, and $Y$.

(b) **[4 pts.]** For a permutation matrix $P \in \mathbb{R}^{L\times L}$, prove that applying the layer to $PX$ produces $PY$.

(c) **[2 pts.]** Explain the consequence for sequence order and state one mathematically precise way to inject positional information.

**3. [9 pts.]** A variational autoencoder uses $p(z) = \mathcal{N}(0, I)$, decoder $p_\theta(x \mid z)$, and encoder $q_\phi(z \mid x)$.

(a) **[4 pts.]** Starting from $\log p_\theta(x)$, derive the evidence lower bound (ELBO) and identify the condition for equality.

(b) **[2 pts.]** Express the exact gap between $\log p_\theta(x)$ and the ELBO as a KL divergence.

(c) **[2 pts.]** If $q_\phi(z \mid x) = \mathcal{N}(\mu_\phi(x), \text{diag}(\sigma_\phi^2(x)))$, give the reparameterized sample used for gradient estimation.

(d) **[1 pts.]** State one mechanism that can cause posterior collapse.

**4. [8 pts.]** For a finite discounted MDP with $\gamma \in (0,1)$, define the Bellman optimality operator

$$(TV)(s) = \max_a \left[ r(s,a) + \gamma \sum_{s'} P(s' \mid s,a) V(s') \right].$$

Prove that $T$ is a $\gamma$-contraction in the supremum norm. Deduce uniqueness of its fixed point and give an explicit error bound for value iteration $V_{k+1} = TV_k$.

## C. Optimization Methods in Artificial Intelligence

Let $f : \mathbb{R}^d \to \mathbb{R}$ be convex and $L$-smooth, and let $R : \mathbb{R}^d \to (-\infty, +\infty]$ be proper, closed, and convex. Assume $F = f + R$ has a minimizer $x^\star$. Define

$$x^+ = \mathrm{prox}_{\gamma R}(x - \gamma \nabla f(x)), \qquad \mathrm{prox}_{\gamma R}(u) = \arg\min_v \left\{ R(v) + \frac{1}{2\gamma} \|v - u\|_2^2 \right\}.$$

**1. [4 pts.]** Derive the optimality condition for $x^+$ and prove that, for every $z \in \mathrm{dom}\, R$,

$$R(x^+) + \langle \nabla f(x), x^+ - z \rangle \le R(z) + \frac{\|x - z\|^2 - \|x^+ - z\|^2 - \|x^+ - x\|^2}{2\gamma}.$$

**2. [6 pts.]** Combine the inequality above with smoothness and convexity to prove

$$F(x^+) - F(z) \le \frac{\|x - z\|^2 - \|x^+ - z\|^2}{2\gamma} - \left( \frac{1}{2\gamma} - \frac{L}{2} \right) \|x^+ - x\|^2.$$

**3. [5 pts.]** For $0 < \gamma \le 1/L$, apply the preceding result to the proximal-gradient iterates $x^{k+1} = \mathrm{prox}_{\gamma R}(x^k - \gamma \nabla f(x^k))$. Prove an $O(1/k)$ bound for $F(x^k) - F(x^\star)$, with the constant shown explicitly.

**4. [6 pts.]** For $R(x) = \lambda \|x\|_1$, derive $\mathrm{prox}_{\gamma R}(u)$ coordinate by coordinate. Your derivation must cover the cases $u_j > \gamma\lambda$, $|u_j| \le \gamma\lambda$, and $u_j < -\gamma\lambda$.

**5. [12 pts.]** Now let $R \equiv 0$ and assume $f$ is $\mu$-strongly convex and $L$-smooth. At iteration $k$, let

$$g_k = \frac{1}{B} \sum_{j=1}^{B} g(x^k, \xi_{k,j}), \qquad \mathbb{E}_k[g(x^k, \xi_{k,j})] = \nabla f(x^k), \qquad \mathbb{E}_k \left\| g(x^k, \xi_{k,j}) - \nabla f(x^k) \right\|^2 \le \sigma^2,$$

where the $B$ samples are conditionally independent, and set $x^{k+1} = x^k - \gamma g_k$.

(a) **[3 pts.]** Prove conditional unbiasedness and the variance bound $\mathbb{E}_k \left\| g_k - \nabla f(x^k) \right\|^2 \le \sigma^2 / B$.

(b) **[5 pts.]** With $\gamma = 2/(L + \mu)$, prove

$$\mathbb{E}_k \left\| x^{k+1} - x^\star \right\|^2 \le q^2 \left\| x^k - x^\star \right\|^2 + \gamma^2 \frac{\sigma^2}{B}, \qquad q = \frac{L - \mu}{L + \mu}.$$

(c) **[4 pts.]** Unroll the recurrence and identify the limiting noise floor. State exactly how increasing $B$ affects this bound.

## D. Natural Language Processing

**1. [7 pts.]** For a center word $w$, a context word $c$, and negative samples $n_1, \dots, n_K$, the Skip-Gram with Negative Sampling objective for one training instance is

$$\ell = -\log \sigma(u_w^\top v_c) - \sum_{j=1}^{K} \log \sigma(-u_{n_j}^\top v_c).$$

Derive the gradients with respect to $v_c$, $u_w$, and every $u_{n_j}$. Explain the update directions and why a semantically valid but sampled "negative" can be harmful.

**2. [9 pts.]** A linear-chain CRF assigns score

$$s_\theta(x, y) = \sum_{t=1}^{T} e_\theta(t, y_t; x) + \sum_{t=2}^{T} A_{y_{t-1}, y_t}$$

and conditional probability $p_\theta(y \mid x) = \exp(s_\theta(x, y))/Z_\theta(x)$ over $K$ labels.

(a) **[3 pts.]** Give a forward recursion for $Z_\theta(x)$ and its time complexity.

(b) **[3 pts.]** When the score is linear in a feature vector $F(x, y)$, derive the gradient of the negative log-likelihood for one labeled sequence.

(c) **[3 pts.]** Prove that the negative log-likelihood is convex in the linear parameter vector. State why this conclusion need not hold when the emission scores are produced by a neural network trained jointly with the CRF.

**3. [7 pts.]** TransE represents a knowledge-base triple $(h, r, t)$ by the score

$$d(h, r, t) = \|e_h + e_r - e_t\|_2^2 .$$

Using corrupted triples $(h', r, t')$, define a margin-ranking training loss and derive the gradients of one active hinge term. State one parameter non-identifiability and one structural failure mode of this translation model.

**4. [10 pts.]** A policy archive contains multiple versions of each document, with validity intervals and source metadata. Design a retrieval-augmented question-answering system that must answer a query "as of" a specified date and attach supporting citations. Specify the offline representation and index, retrieval and reranking computation, temporal consistency rule, generator input and training objectives, inference sequence, dominant complexity terms, and a leakage-free evaluation protocol. Include at least two distinct failure modes and corresponding diagnostics.